Forecasting and Macroeconomic Analytics

ECON 522

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Forecasting

General Considerations for forecasting

Let's start by highlighting, in no particular order, a number of considerations relevant for any forecasting task.

1. (**Forecast Object**) What is the object that we want to forecast? Is it a time series, event outcome or event timing, or other? Strategies depend on the nature of the object being forecast.
2. (**Information Set**) On what information will the forecast be based? Are we forecasting one series, several, or thousands? What the quantity and quality of the data? Mostly time series itself.
3. (**Model Uncertainty and Improvement**) Does our forecasting model match the true DGP? Of course not. All models are false: they are intentional abstractions of a much more complex reality. A model might be useful for certain purposes and poor for others.

General Considerations for forecasting

h = Forecast horizon

4. (**Forecast Horizon**) What is the forecast horizon of interest, and what determines it? one month ahead, one year ahead, or ten years ahead (called h-step-ahead forecasts, in this case for $h = 1, h = 12$ and $h = 120$ months)? Appropriate forecasting strategies likely vary with the horizon.

ex. is COVID a structural change? or just a blip?

5. (**Structural Change**) Are the approximations to reality that we use for forecasting (i.e., our models) stable over time? Generally not. Things can change for a variety of reasons, gradually or abruptly: we need methods of detecting and adapting to structural change.
6. (**Forecast Statement**) How will our forecasts be stated? If, for example, the object to be forecast is a time series, are we interested in a single "best guess" forecast, a "reasonable range" of possible future values that reflects the underlying uncertainty associated with the forecasting problem, or a full probability distribution of possible future values? What are the associated costs and benefits?

General Considerations for forecasting

7. (**Forecast Presentation**) How best to present forecasts? Except in the simplest cases, like a single h-step-ahead point forecast, graphical methods are valuable, not only for forecast presentation but also for forecast construction and evaluation.
8. (**Decision Environment and Loss Function**) What is the decision environment in which the forecast will be used? In particular, what decision will the forecast guide? How do we quantify what we mean by a "good" forecast, and in particular, the cost or loss associated with forecast errors of various signs and sizes?

General Considerations for forecasting

9. (**Model Complexity and the Parsimony Principle**) What sorts of models, in terms of complexity, tend to do best for forecasting in business, finance, economics, and government? Bigger forecasting models are not necessarily better, and indeed. Remember the "parsimony principle".
10. (**Unobserved Components**) In the leading time case of time series, have we successfully modeled trend? Seasonality? Cycles? Some series have all such components, and some not. They are driven by very different factors, and each should be given serious attention.

Forecast Horizon

PSEUDO OOS

1970 ⟶ 2024

T+h $\xrightarrow{2025}$

use information from just 1970-2024 to forecast post-2024. Then use existing 2024-2026 data to confirm forecasting performance

OUT OF SAMPLE

Feb, 2026 ⟶ future

Test model using existing data, then wait and see if your forecast was accurate.

Horizon

1. h-step ahead
2. Out of sample versus in sample versus pseudo out of sample
3. Nowcasting and Backcasting:

Going back and fill the missing piecing

prediction for NOW, present, using existing data.

↳ ex) THIS month, but we don't data for THIS month yet.

A single point in the time series

Best vs. worst possible case scenario range

4. Point forecast, density forecast, interval forecast

distribution for forecasting: what's the range of the possible outcomes in forecasting, and distribution across this range?

↳ Probability of ending up anywhere between best & worst case scenario

5. For event, probability forecast

probability of a specific event occuring.

The Loss Function

Let y denote a series and $\hat{y}$ its forecast. The corresponding forecast error, e, is the difference between the realization and the previously-made forecast,

$$e = y - \hat{y}$$

We consider loss functions of the form $L(e)$. This means that the loss associated with a forecast depends only on the size of the forecast error. We might require the loss function $L(e)$ to satisfy three conditions:

1. $L(0) = 0$. That is, no loss is incurred when the forecast error is zero.
2. $L(e)$ is **continuous**. That is, nearly-identical forecast errors should produce nearly-identical losses.
3. $L(e)$ is **increasing** on each side of the origin. That is, the bigger the absolute value of the error, the bigger the loss.

Apart from these three requirements, we impose no restrictions on the form of the loss function.

Symmetric Loss Functions

OLS minimizes $L(e)$ when $L(e) = \sum e_t^2$

Quadratic loss is given by

$$L(e) = e^2,$$

· heavily penalizes big errors
· mathematically easy to minimize

- It is a reasonable approximation in most cases (why?)
- It is easy to compute and mathematical convenient → minimize by FOC = 0

Another important symmetric loss function is **absolute loss**, or absolute error loss, given by

$$L(e) = |e|.$$

- Similar properties yet different. Why?

Asymmetric Loss Functions

An important asymmetric loss function is **"linlin loss"** (linear on each side of the origin, with generally different slopes), given by

$$L(e) = \begin{cases} a|e|, & \text{if } e > 0 \\ b|e|, & \text{if } e \leq 0 \end{cases}$$

- Can you give me an example of when you would use this loss function?

You can also have a **state dependent loss.** For example a direction of change forecast that only penalizes you if you get the direction wrong.

Loos functions comparison

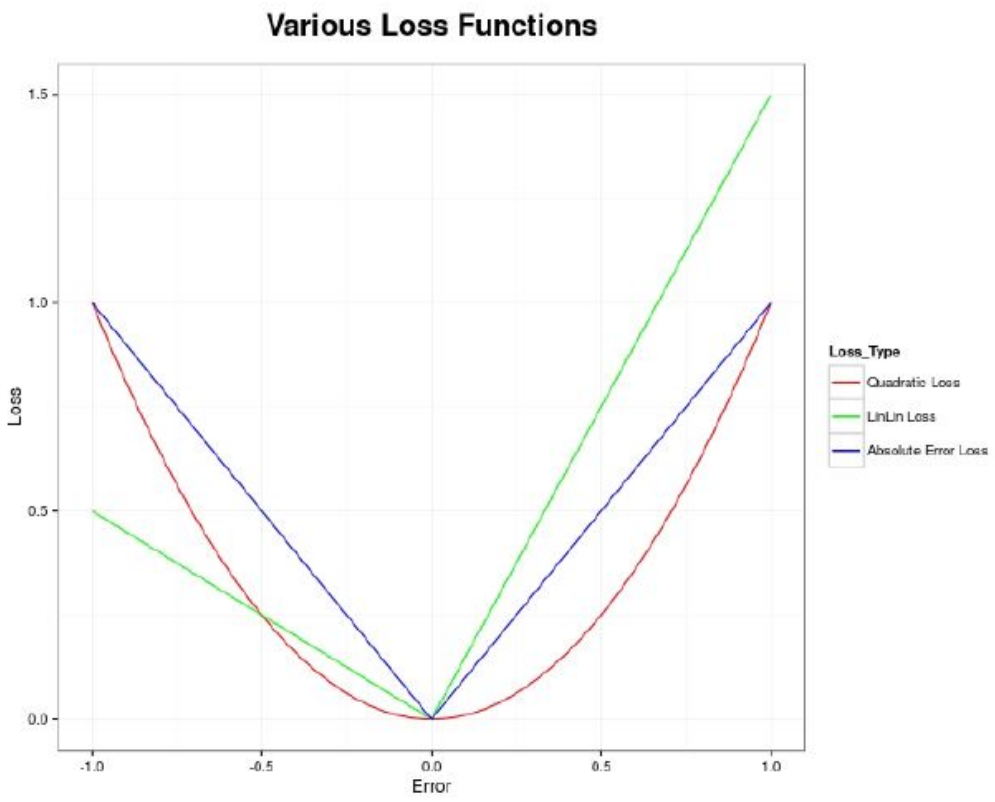


Figure 2.5: Quadratic, Absolute, and Linlin Loss Functions

Optimal Forecast

What is the optimal forecast? The forecast with the **smallest conditional expected loss**.

$$\hat{y}(x)^* = \text{argmin}_{\hat{y}(x)} \iint L(y - \hat{y}(x)) f(y, x) dy dx.$$

$\min_{\hat{y}} E[L(e)]$

find forecast $\hat{y}$ that minimizes $E[L(e)]$ of random variable error e.

- Under quadratic loss, the optimal forecast is the *conditional mean*: $\hat{y}(x)^* = E(y \mid x)$. x could be lagged y.
- Under absolute loss, the optimal forecast is the *conditional median*, $\hat{y}(x)^* = Q_{.50 \cdot 100\%}(y \mid x)$
- Under lin-lin loss, the optimal forecast is the *conditional* $d \cdot 100\%$ *quantile*, where $d = \frac{b}{a+b} = \frac{1}{1+a/b}$. That is, $\hat{y}(x)^* = Q_{d \cdot 100\%}(y \mid x)$.

Under asymmetric $L(e)$ loss (e.g., linlin), optimal forecasts are biased, whereas the conditional mean forecast is unbiased. Bias is optimal under asymmetric loss because we can gain on average by pushing the forecasts in the direction such that we make relatively few errors of the more costly sign.

Forecasting Trend

Deterministic trend is easy to forecast.

$$\hat{y}_{T+h,T} = \hat{\beta}_0 + \hat{\beta}_1 TIME_{T+h}.$$

- Note the notation. Each book will have a different notation.
- What would the density forecast be? $N\left(\hat{y}_{T+h,T}, \hat{\sigma}^2\right)$ Why?
- What would the interval forecast be? $\hat{y}_{T+h,T} \pm 1.96\hat{\sigma}$,

What is the best forecast of a variable with stochastic trend?

A good example is in FD section 5.2.4 which you will repeat in your lab

ex) Quadratic Loss :

$$E(y \mid x)$$

for us, linear model (OLS)

$$y_t = \beta_0 + \beta_1 t + \varepsilon_t \qquad t = \overbrace{1, 2, \dots,}^{\text{past}} T \leftarrow \text{Now}$$

where I start.

OOS forecast : $\hat{y}_{T+h,T} = \hat{\beta}_0 + \hat{\beta}_1 (T+h) + 0$

Value of data @ time T + horizon h

↳ when in future we want to forecast

$$\sim N(\hat{y}_{T+h,T}, \hat{\sigma}^2)$$

$$CI = \hat{y}_{T+h,T} \pm 1.96\,\hat{\sigma}$$

Forecasting Seasonality

Seasonality is also deterministic so it is easy to forecast:

$$\hat{y}_{T+h,T} = \sum_{i=1}^{s} \hat{\gamma}_i D_{i,T+h}.$$

1. Density and interval forecasts are exactly as in the trend model. $N\left(\hat{y}_{T+h,T}, \hat{\sigma}^2\right)$, and $\hat{y}_{T+h,T} \pm 1.96\hat{\sigma}$.

A good example is FB section 5.3.4 which you may also repeat in the lab.

Forecasting With ARMA models - Information Set

- We denote the time- T information set by Ω_T. It seems most natural to think of the information set as containing the available past history of the series,

All the info I know @ time T.

$$\Omega_T = \{y_T, y_{T-1}, y_{T-2}, \ldots\}$$

if I have other variable(s) $x_T, x_{T-1}, x_{T-2}, \ldots\}$

where we imagine history as having begun in the infinite past.

- So long as y is covariance stationary, however, we can just as easily express the information available at time T in terms of current and past shocks,

$$\Omega_T = \{\varepsilon_T, \varepsilon_{T-1}, \varepsilon_{T-2}, \ldots\}$$

as we can get figure out the current and part ε from the current and lagged y.

- So we often say

$$\Omega_T = y_T, y_{T-1}, y_{T-2}, \ldots, \varepsilon_T, \varepsilon_{T-1}, \varepsilon_{T-2}, \ldots$$

Forecasting With ARMA models - Linear Forecast

- Based upon that information set, we want to find the optimal forecast of y at some future time $T + h$.
- The optimal forecast is the the forecast that minimizes expected loss.
- It turns out that under reasonably weak conditions the optimal forecast is the conditional mean,

$$E\left(y_{T+h} \mid \Omega_T\right)$$

- In general, the conditional mean need not be a linear function of the elements of the information set. Because linear functions are particularly tractable, we prefer to work with linear forecasts (linear least square forecast).
- In the Gaussian case the conditional expectation is exactly linear, so that

$$E\left(y_{T+h} \mid \Omega_T\right) = P\left(y_{T+h} \mid \Omega_T\right).$$

so we can work with the ARMA models

Forecasting With ARMA models - Finite MA(2)

Consider an $MA(2)$ process,

$$y_t = \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2}$$
$$\varepsilon_t \sim WN\left(0, \sigma^2\right)$$

Suppose we're standing at time T and we want to forecast y_{T+1}. First we write out the process for $T+1$,

$$y_{T+1} = \varepsilon_{T+1} + \theta_1 \varepsilon_T + \theta_2 \varepsilon_{T-1}$$

Thus

$$y_{T+1,T} = P\left(y_{T+1} \mid \Omega_T\right) = \theta_1 \varepsilon_T + \theta_2 \varepsilon_{T-1}.$$

$$y_{T+2,T} = \theta_2 \varepsilon_T$$

$$y_{T+h,T} = 0$$

for all $h > 2$.

example on prev. page:

* Any ε W.N. in the future, you cannot forecast

constant

$$MA(2): \quad y_t = \mu + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2}$$

$$\hat{y}_{T+h,T}$$

can't forecast W.N. into future.

Can forecast W.N. of present & past.

$$h=1 \Rightarrow y_{T+1} = \mu + \varepsilon_{T+1} + \theta_1 \varepsilon_T + \theta_2 \varepsilon_{T-1}$$

$$\hat{y}_{T+1,T} = \hat{\mu} + 0 + \hat{\theta}_1 \hat{\varepsilon}_T + \theta_2 \hat{\varepsilon}_{T-1}$$

Forecast error: $e_{T+1,T} = y_{T+1} - y_{T+1,T} = \varepsilon_{T+1} \Rightarrow WN$

$$h=2 \Rightarrow y_{T+2} = \mu + \varepsilon_{T+2} + \theta_1 \varepsilon_{T+1} + \theta_2 \varepsilon_T$$

$$\hat{y}_{T+2,T} = \hat{\mu} + 0 + 0 + \hat{\theta}_2 \hat{\varepsilon}_T$$

Forecast error: $e_{T+2,T} = y_{T+2} - y_{T+2,T} = \varepsilon_{T+2} + \theta_1 \varepsilon_{T+1} \Rightarrow MA(1)$

$$h=3 \Rightarrow y_{T+3} = \mu + \varepsilon_{T+3} + \theta_1 \varepsilon_{T+2} + \theta_2 \varepsilon_{T+1}$$

$$\hat{y}_{T+3,T} = \hat{\mu} + 0 + 0 + 0$$

Forecast error for any h: $e_{T+h,T} = y_{T+h} - y_{T+h,T} = \varepsilon_{T+h} + \theta_1 \varepsilon_{T+h-1} + \theta_2 \varepsilon_{T+h-2}$

Forecast error for MA(2)

Now let's compute the corresponding forecast errors. We have:

$$e_{T+1,T} = \varepsilon_{T+1} WN$$
$$e_{T+2,T} = \varepsilon_{T+2} + \theta_1 \varepsilon_{T+1} (MA(1))$$
$$e_{T+h,T} = \varepsilon_{T+h} + \theta_1 \varepsilon_{T+h-1} + \theta_2 \varepsilon_{T+h-2} (MA(2)),$$

Forecast error variance for MA(2)

Finally, the forecast error variances are:

$$\sigma_1^2 = \sigma^2$$
$$\sigma_2^2 = \sigma^2 \left(1 + \theta_1^2\right)$$
$$\sigma_h^2 = \sigma^2 \left(1 + \theta_1^2 + \theta_2^2\right),$$

for all $h > 2$. Moreover, the forecast error variance for $\mathrm{h} > 2$ is just the unconditional variance of y_t.

Forecasting with MA(q)

Now consider the general $MA(q)$ case. The model is

we always start with MA because any model can be written as a moving average.

$$y_t = \varepsilon_t + \theta_1 \varepsilon_{t-1} + \ldots \theta_q \varepsilon_{t-q}$$

$$y_{T+h,T} = \theta_h \varepsilon_T + \theta_{h-1)} \varepsilon_{T-1} + \ldots .$$

$$e_{T+h,T} = \varepsilon_{T+h} + \theta_1 \varepsilon_{T+h-1} + \ldots + \theta_{h-1} \varepsilon_{T+1}$$

- Thus, an $MA(q)$ process is not forecastable (apart from the unconditional mean) more than q steps ahead.
- Forecast error is such that $e_{T+h,T} = MA(h-1)$ for $h \leq q$ and $e_{T+h,T} = MA(q)$ for $h > q$. The h-step-ahead forecast error for $h > q$ is just the process itself, minus its mean.
- Finally, consider the forecast error variances. For $h \leq q$,$\sigma_h^2 \leq \text{var}(y_t)$, whereas for $h > q$, $\sigma_h^2 = \text{var}(y_t)$.

Forecasts Properties

- For forecasts to be optimal the forecast error should have mean zero and be uncorrelated with $\varepsilon_t, \varepsilon_{t-1}, \ldots$. Check that this is true.
- In addition, for optimal forecast we should have $\mathrm{COV}\left(Y_{T+h,T}, e_{t+h}\right) = 0$.
- $MSE = E\left(Y_{T+h} - Y_{T+h,T}\right)^2 = \left(1 + \theta_1^2 + \theta_2^2 + \ldots. + \theta_{h-1}^2\right)\sigma^2$
- $Y_{tT+h} = Y_{T+h,T} + e_{T,h}$ so $\mathrm{VAR}\left(Y_{T+h}\right) = \mathrm{VAR}\left(Y_{T+h,T}\right) + \mathrm{VAR}\left(e_{T,h}\right) + 2\,\mathrm{COV}\left(Y_{T+h,T}, e_{T,h}\right) = \mathrm{VAR}\left(Y_{T+h,T}\right) + \mathrm{VAR}\left(e_{T,h}\right)$,
- $\mathrm{VAR}\left(Y_{T+h}\right) > \mathrm{VAR}\left(Y_{T+h,T}\right)$. This carefully about what this means for when you graph actually and forecasts.
- Updating rule $Y_{T+h,T} = Y_{T+h+1,T-1} + \theta_h \varepsilon_t$
- Updating rule for forecast errors is $e_{T,h} = e_{T1,h+1} - \theta_h \varepsilon_t$

$e_{T,h} \Leftrightarrow e_{T+h,T}$ notation-wise

we can always write

$$y_{T+h} = \hat{y}_{T+h,T} + e_{T,h}$$

good forecast means uncorrelated, $\mathrm{cov}(y_{T+h,T}, e_{T+h}) = 0$

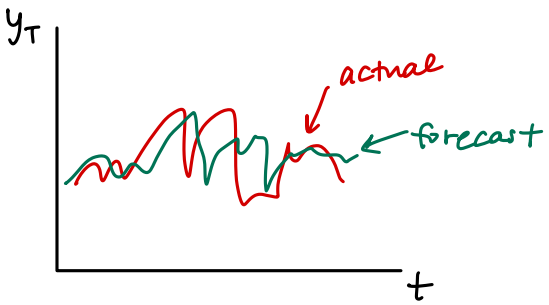
$$\mathrm{VAR}(y_{T+h}) = \mathrm{VAR}(\hat{y}_{T+h,T}) + \mathrm{VAR}(e_{T,h})$$

$\downarrow$ Therefore

$$\mathrm{VAR}(y_{T+h}) > \mathrm{VAR}(\hat{y}_{T+h,T})$$

$\downarrow$ Thus

don't expect forecast to match up identically with actual, as variance will always be smaller.



- Actual $\neq$ forecast exactly. ALWAYS.
- forecast will be a bit smoother, with a SMALLER variance.

Interval and Density Forecast - MA

The definition of the h-step-ahead forecast error is $e_{T+h,T} = y_{T+h} - y_{T+h,T}$. Equivalently, the h-step-ahead realized value, y_{T+h}, equals the forecast plus the error,

$$y_{T+h} = y_{T+h,T} + e_{T+h,T}.$$

- If the innovations are normally distributed, then the future value of the series of interest is also normally distributed conditional upon the information set available at the time the forecast was made, and so we have the 95%h-step ahead interval forecast $y_{T+h,T} \pm 1.96\sigma_h$.[12]
- The h-step-ahead density forecast as

$$N\left(y_{T+h,T}, \sigma_h^2\right)$$

Interval Estimation for MA(2)

Consider the MA(2)

$$y_t = \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2}$$
$$\varepsilon_t \sim WN\left(0, \sigma^2\right).$$

Assuming normality, the 1-step-ahead 95% interval forecast is

$$y_{T+1,T} = \left(\theta_1 \varepsilon_T + \theta_2 \varepsilon_{T-1}\right) \pm 1.96\sigma$$

and the 1-step-ahead density forecast is

$$N\left(\theta_1 \varepsilon_T + \theta_2 \varepsilon_{T-1}, \sigma^2\right)$$

Operationally you replace population parameters with estimated one.

Interval Estimation for MA(2)

$$y_{T+2,T} = \theta_2 \varepsilon_T,$$

with corresponding forecast error

$$e_{T+2,T} = \varepsilon_{T+2} + \theta_1 \varepsilon_{T+1},$$

and forecast-error variance

$$\sigma_2^2 = \sigma^2 \left(1 + \theta_1^2\right)$$

Then, if desired, we can construct operational 2-step-ahead interval and density forecasts, as

$$\hat{y}_{T+2,T} \pm z_{\alpha/2} \hat{\sigma}_2$$

and

$$N\left(\hat{y}_{T+2,T}, \hat{\sigma}_2^2\right)$$

Side note to be precise

The exact forecast error is

$$\hat{e}_{T+2,T} = y_{T+2} - \hat{y}_{T+2,T} = \varepsilon_{T+2} + \theta_1 \varepsilon_{T+1} + \left(\theta_2 - \hat{\theta}_2\right) \varepsilon_T,$$

the variance of which is very difficult to evaluate. So we make a convenient approximation: we ignore parameter estimation uncertainty by assuming that estimated parameters equal true parameters. We therefore set

$$\left(\theta_2 - \hat{\theta}_2\right)$$

to zero, which yields

$$\hat{e}_{T+2,T} = \varepsilon_{T+2} + \theta_1 \varepsilon_{T+1}$$

with variance

$$\sigma_2^2 = \sigma^2 \left(1 + \theta_1^2\right)$$

Forecasting With ARMA Models - AR(1)

Since we can write a AR(1) as a MA of infinite order, we could simply consider all we have learned from MA here. However, it turns out that thinking in terms of AR model is much simpler for forecasting. Consider the AR(1)

$$y_t = \phi y_{t-1} + \varepsilon_t$$
$$\varepsilon_t \sim WN\left(0, \sigma^2\right)$$

To construct the 1-step-ahead forecast, we write out the process for time $T+1$, $y_{T+1} = \phi y_T + \varepsilon_{T+1}$

$$y_{T+1,T} = \phi y_T$$

Now let's construct the 2-step-ahead forecast. Write out the process for time $T+2$, $y_{T+2} = \phi y_{T+1} + \varepsilon_{T+2}$. Then project directly on the time- T information set to get

$$y_{T+2,T} = \phi y_{T+1,T}$$

and so on $y_{T+h,T} = \phi y_{T+h-1,T}$

<u>AR(1)</u> $y_t = \phi y_{t-1} + \varepsilon_t$

$$y_{T+1} = \phi y_T + \varepsilon_{T+1}$$

$$y_{T+1,T} = \hat{\phi} y_T$$

$$y_{T+2} = \phi y_{T+1} + \varepsilon_{T+2}$$

$$y_{T+2,T} = \hat{\phi} y_{T+1} = \hat{\phi} y_{T+1,T} = \hat{\phi}^2 y_T$$

$$y_{T+h,T} = \hat{\phi}^h y_T = \hat{\phi} \cdot y_{T+h-1,T}$$

Forecasting With ARMA models - ARMA

Forecasting with ARMA is a combination of what we have seen. For example, for ARMA(1,1)

$$y_{T+1} = \phi y_T + \varepsilon_{T+1} + \theta \varepsilon_T$$

$$y_{T+1,T} = \phi y_T + \theta \varepsilon_T.$$

$$y_{T+2} = \phi y_{T+1} + \varepsilon_{T+2} + \theta \varepsilon_{T+1} = \phi^2 y_T + \phi \theta \varepsilon_T.$$

$$y_{T+2,T} = \phi y_{T+1,T}$$

Clearly, for $h > 1$

$$y_{T+h,T} = \phi y_{T+h-1,T},$$

ARMA preferred because
you can forecast far into future

Forecasting With ARMA models - Interval and Density Forecast

It is similar to what we saw before. As an example consider ARMA(1,1) $y_t = \phi y_{t-1} + \varepsilon_t + \theta\varepsilon_{t-1}$ Substitute backward for y_{t-1} to get

$$\begin{aligned} y_t &= \phi\left(\phi y_{t-2} + \varepsilon_{t-1} + \theta\varepsilon_{t-2}\right) + \varepsilon_t + \theta\varepsilon_{t-1} \\ &= \varepsilon_t + (\phi+\theta)\varepsilon_{t-1} + \ldots \end{aligned}$$

We need not substitute back any farther, because the 2-step-ahead forecast error variance is

$$\sigma_2^2 = \sigma^2\left(1 + b_1^2\right)$$

where b_1 is the coefficient on ε_{t-1} in the moving average representation of the ARMA $(1,1)$ process, which we just calculated to be $(\phi+\theta)$. Thus the 2 -stepahead interval forecast is $y_{T+2,T} \pm 1.96\sigma_2$, or $\left(\phi^2 y_T + \phi\theta\varepsilon_T\right) \pm 1.96\sigma\sqrt{1+(\phi+\theta)^2}$. We make this operational as $\left(\hat{\phi}^2 y_T + \hat{\phi}\hat{\theta}\varepsilon_T\right) \pm 1.96\hat{\sigma}\sqrt{1+(\hat{\phi}+\hat{\theta})^2}$.